

**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2008
Higher 1**

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CANDIDATE NAME

CLASS

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BIOLOGY

8875/01

Paper 1 Multiple Choice

19 September 2008

1 hour

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

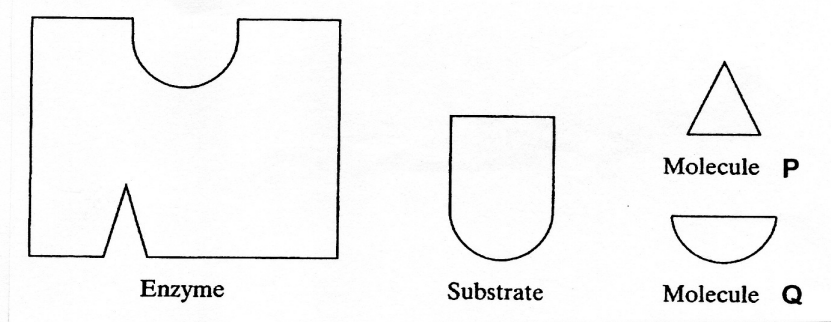
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet

This paper consists of **12** printed pages excluding this cover page.

Attempt **all** the questions.

- Monosaccharide generally have a molecular formula
 - that includes a $-SH$ group
 - in which hydrogen, carbon and oxygen are present in the ratio of 2:1:1
 - that includes at least one hydrocarbon tail
 - in which carbon, hydrogen and oxygen are present in a 2:1:2 ratio
- Which of the following is true of **lipids**?
 - Products of its hydrolysis can pass freely through the bilayer.
 - Products of its hydrolysis always include fatty acids and glycerol.
 - Lipids are involved in stabilizing the lipid bilayer.
 - Saturation of fatty acids causes an increase in membrane fluidity.
- The R group or side chain of the amino acid serine is CH_2-OH . The R group of the amino acid alanine is $-CH_3$. Where would you expect to find these amino acids in a globular protein in aqueous solution?
 - Both serine and alanine would be in the interior and on the exterior of the globular protein.
 - Serine would be in the interior, and alanine would be on the exterior of the globular protein.
 - Both serine and alanine would be in the interior of the globular protein.
 - Alanine would be in the interior, and serine would be on the exterior of the globular protein.
- The diagram shows an enzyme, its substrate and two other molecules P and Q. Which of the following statements best describes the possible role of P and Q?

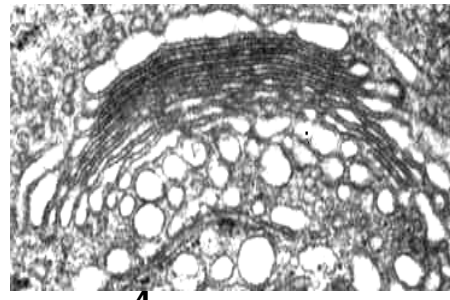


- Molecule P is an activator that binds to the allosteric site of the enzyme while molecule Q has no effect on enzyme action.
 - Molecule P is a competitive inhibitor that binds to the allosteric site of the enzyme while molecule Q is a non-competitive inhibitor that binds to the active site of the enzyme.
 - Molecule P is an inhibitor that binds to the allosteric site of the enzyme while molecule Q is a competitive inhibitor that binds to the active site of the enzyme.
 - Both molecule P and molecule Q have no effect on enzyme action.
- The following contains images of a number of structures that can be seen in a cell.

1

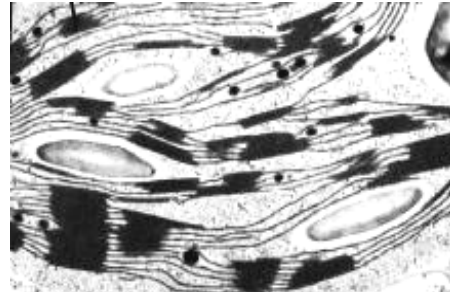
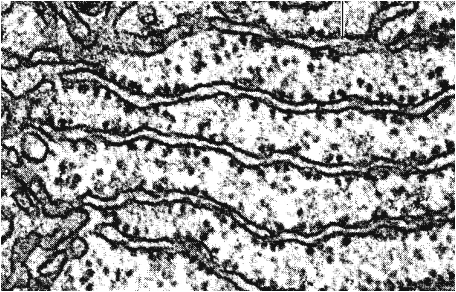


2



4

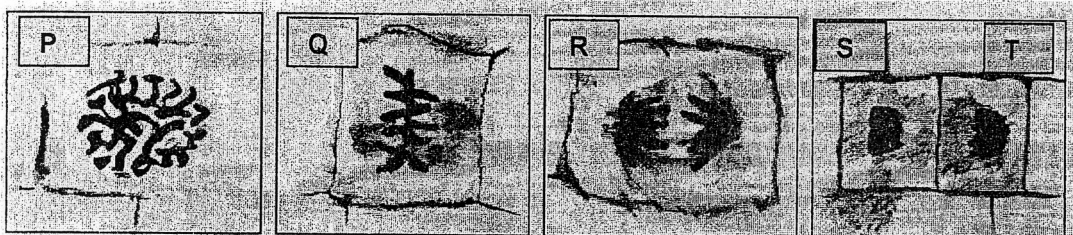
3



Identify structures 1 – 4.

	1	2	3	4
A.	chloroplast	nucleus	ribosome	centriole
B.	centriole	Golgi apparatus	rough endoplasmic reticulum	chloroplast
C.	centriole	Golgi apparatus	rough endoplasmic reticulum	smooth endoplasmic reticulum
D.	flagellum	Smooth endoplasmic reticulum	Golgi apparatus	chloroplast

6. Which of the following is/are the most likely consequence for a cell lacking functional Golgi bodies?
1. The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
 2. The cell dies from a lack of enzymes to digest food taken in by endocytosis.
 3. The cell dies from the accumulation of worn-out organelles within itself.
 4. The cell is unable to reproduce itself
 5. The cell is unable to export its enzymes or peptide hormones.
- A. 1 and 5
 B. 2 and 3
 C. 3, 4 and 5
 D. 1,2, 3 and 5
7. Refer to the following diagrams in the figure below, which shows the appearance of cells from a plant tissue. Cells P, Q and R are undergoing various stages of mitosis. Cells S and T are daughter cells formed immediately after cell division.



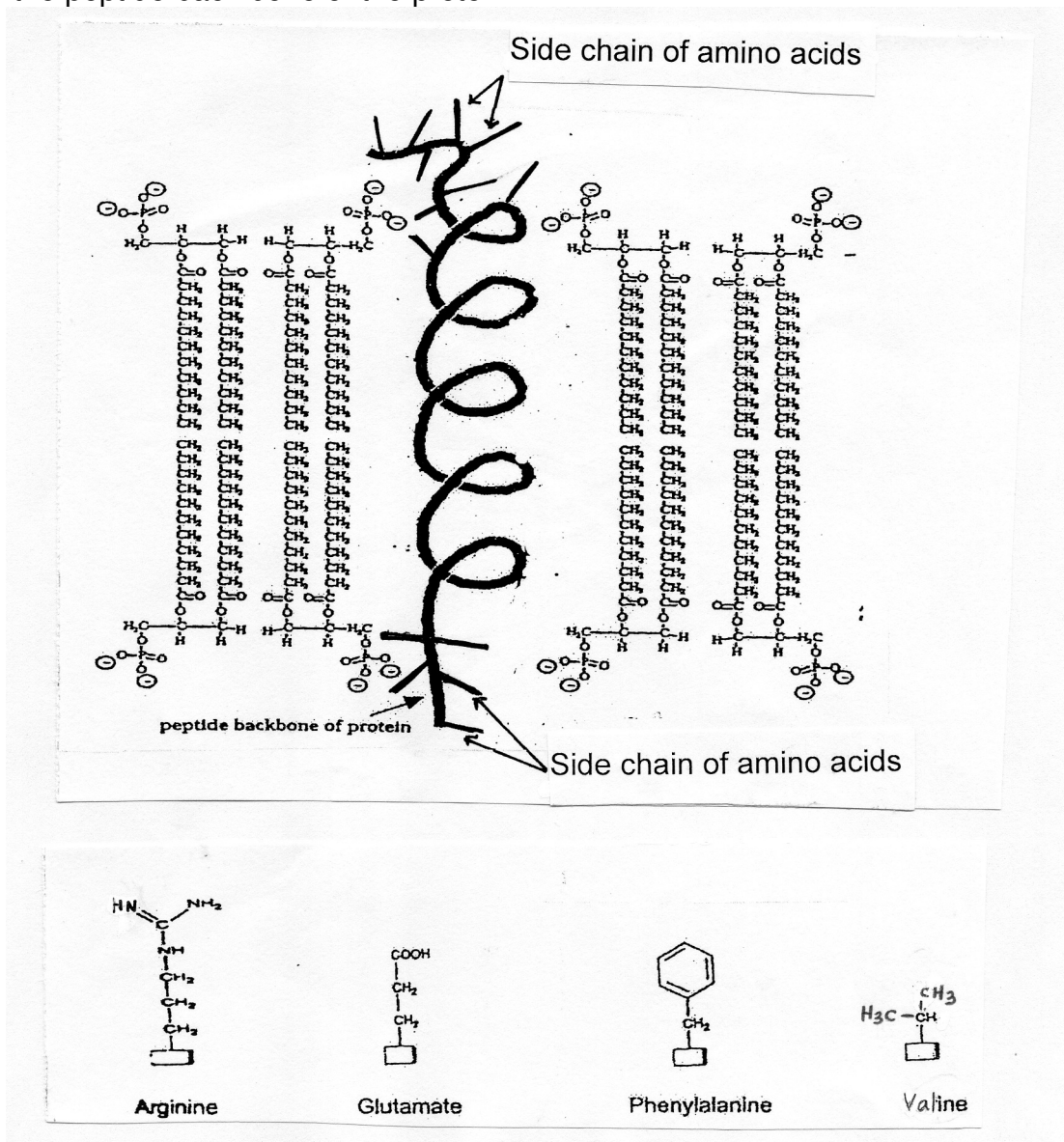
The plant tissue had been grown in a culture medium containing radioactive DNA nucleotides. The level of radioactivity in each cell was measured and the results are shown in the table below:

Cell in Diagram	Level of radioactivity (in arbitrary unit)
P	102
Q	100
R	52
S	51

Cells with the same amount of DNA have approximately the same level of radioactivity. For which cell was the level of radioactivity INCORRECTLY measured?

- A. P
 B. Q
 C. R
 D. S

8. With reference to the figure below, which one of the following amino acids, with only the side chains indicated, is likely to be found at any of the positions protruding from the peptide back bone of the protein



Arginine and Glutamate

- B. Glutamate and Phenylalanine
C. Phenylalanine and Valine
D. Arginine and Valine

9. A biochemist isolated and purified a small amount of DNA a few hundred base pairs long from the bones of a dinosaur fossil. He decided to amplify the amount of the DNA sample using polymerase chain reaction (PCR). When the PCR reagent mixture was added into the thermocycler, no PCR took place. Which of the following was most likely to have resulted in this failure?
- A. Taq DNA polymerase that was pre-treated at 35°C.
B. Free DNA nucleotides with modified 3' ends were added.
C. 2 short DNA sequences to function as primers were added.
D. Primers were added in excess to the PCR mixture.
10. A mutation involving the substitution of one nitrogenous base for another has altered the base sequence of a DNA molecule as shown below:

Normal **A G C A T G G A T C C T**

Mutant **A G C A T G C A T C C T**

The table shows six codons and the corresponding amino acids into which each is translated.

mRNA codon	Amino acid
AAG	Lysine
CUA	Leucine
GGA	Glycine
GUA	Valine
UAC	Tyrosine
UCG	serine

The mutation has changed the amino acid

- A. leucine to valine
- B. lysine to glycine
- C. serine to leucine
- D. valine to serine

11. A synthetic peptide contains ten amino acids of four different kinds. What is the theoretical minimum number of tRNA molecules required to translate the mRNA for this synthetic peptide?

- A. 4
- B. 10
- C. 12
- D. 40

12. Which of the following would most likely to lead to cancer?

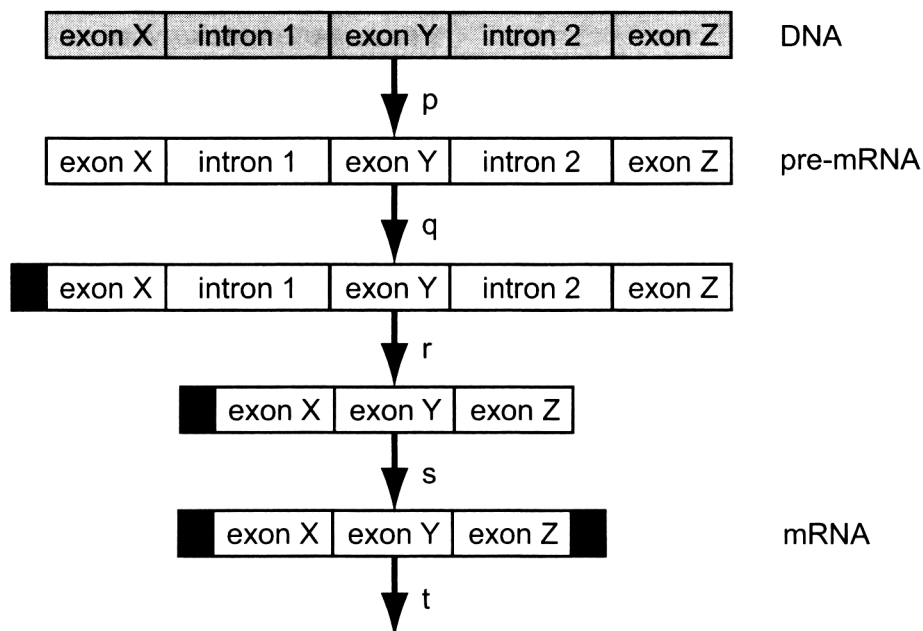
- A. Hyperactivity of a proto-oncogene and activation of a tumour-suppressor gene.
- B. The failure of both a proto-oncogene and a tumour-suppressor gene to produce proteins.
- C. Hyperactivity of a proto-oncogene and inactivation of a tumour-suppressor gene.
- D. Failure of a proto-oncogene to produce a protein and multiplication of a tumour-suppressor gene.

13. Which of the following statement about chromosome structure is true?

- 1 The 30 nm chromatin fibre has a solenoid structure.
- 2 Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
- 3 Euchromatin is the more diffuse region of the interphase chromosome and is transcriptionally inactive.
- 4 The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold.

- A. 1 and 2
- B. 1 and 4
- C. 2 and 3
- D. 3 and 4

14. The diagram represents the processing of pre-messenger RNA.



During the process of pre-messenger RNA each of the listed events occurs.

- 1 capping
- 2 polyadenylation
- 3 splicing
- 4 transcription
- 5 translation

Which correctly identifies the processes p, q, r, s and t?

	<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>	<i>t</i>
A.	5	1	3	2	4
B.	4	1	3	2	5
C.	5	2	1	3	4
D.	4	2	1	3	5

15. Which of the following statements about the eukaryotic control elements are **CORRECT**?

- (i) Attachment of RNA polymerase at the promoter is achieved with the help of a series of transcription factors.
- (ii) Distal control elements may increase the basal activity of the promoter.
- (iii) A given gene may have multiple enhancers, each active at different stages of cell development, cell location or cell type.
- (iv) Repressors bind to silencers to repress transcription by preventing RNA polymerase from binding to the promoter.

- A. (i) and (iv)
- B. (ii) and (iii)
- C. (ii), (iii) and (iv)
- D. (i), (ii), (iii) and (iv)

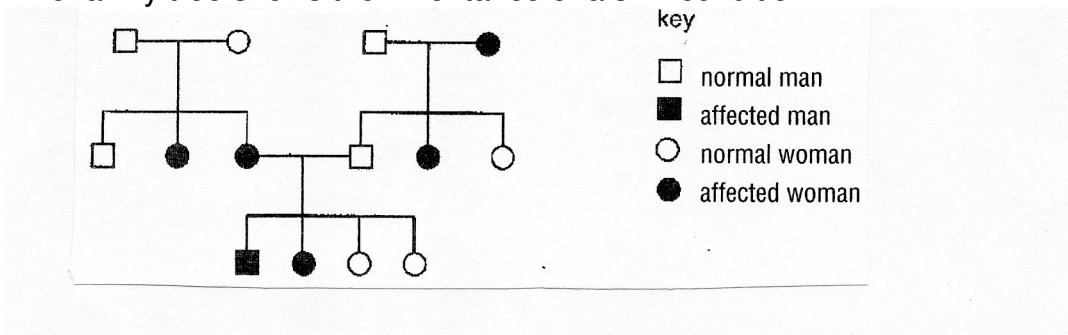
16. Which of the following provides a possible mechanism for the production of a wide number of types of antibody proteins from a small number of genes?

- A. Presence of control elements to regulate gene expression.
- B. Alternative splicing
- C. Crossing over and random segregation
- D. Gene deletions

17. A phenotype ratio of 9:3:3:1 in the offspring of a mating of two organisms heterozygous for two traits is expected when

- A. The two genes are located on the same chromosome.
- B. The two genes are located on different chromosomes.
- C. Only recessive traits are scored.
- D. The alleles are codominant.

18. The family tree shows the inheritance of a skin condition.



What is the most likely genetic basis for the occurrence of this defect?

- A. An autosomal recessive
- B. An autosomal dominant
- C. Sex-linked recessive
- D. Sex-linked dominant

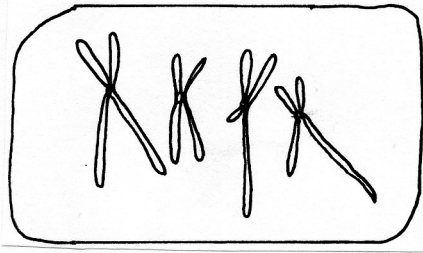
19. There are 3 alleles at the N locus that determines feather colour in dwarf chicken. The allele N^R codes for red is dominant over the other alleles, N and n. The allele N codes for dusky red and is dominant over the allele n, which codes for white.

A red and a dusky red chicken were mated and produced offspring that have red feathers or dusky red feathers. In another cross, one of the dusky red feathered offspring was mated with a white feathered chicken to produce young chicken with a

ratio of 1 dusky red to 1 white. Which of the following is likely to be the parental genotype of the first cross?

- A. $N^R N \times NN$
- B. $N^R N \times Nn$
- C. $N^R n \times NN$
- D. $N^R n \times nn$

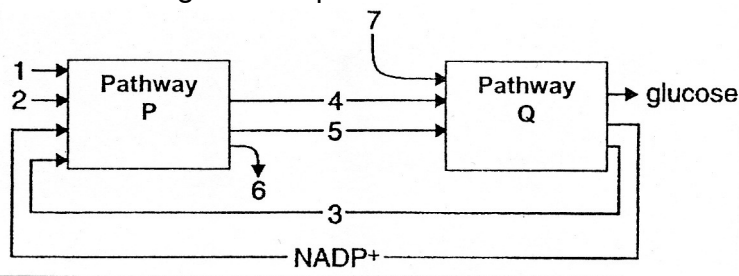
20. The diagram below shows the chromosomes from a diploid cell.



Which of the following could have led to the state of the chromosomes in this cell?

- A. Inversion of a segment of chromosome.
- B. Deletion of a segment of a chromosome.
- C. Duplication of a segment of a chromosome.
- D. Translocation of a segment of a chromosome.

Refer to the diagram for question 21 and 22.



The diagram demonstrates the relationship between the two biochemical pathways in photosynthesis. The arrows could represent reactants or products.

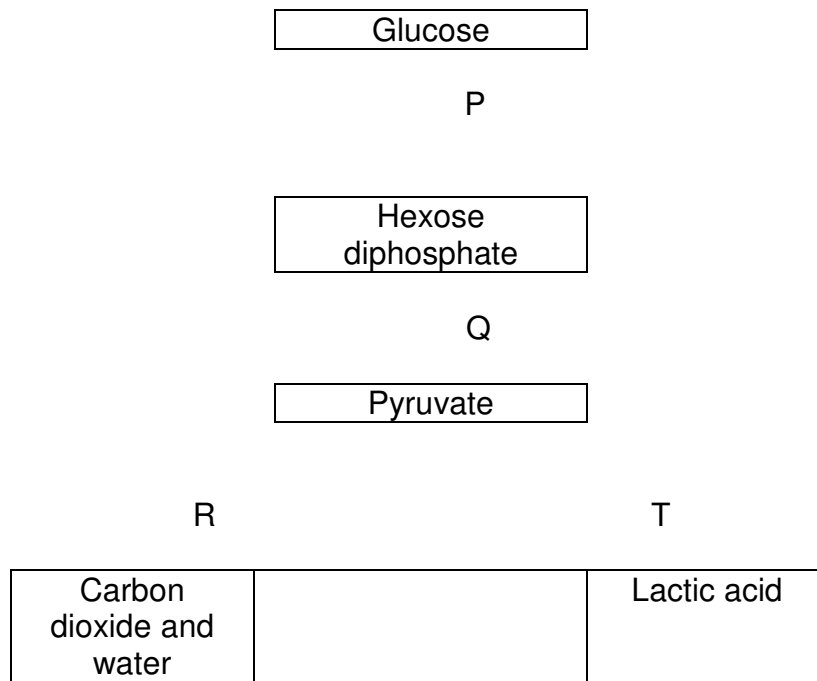
21. Which combination of reactants/ products best represents arrows 1, 4, 6 and 7?

	Arrow 1	Arrow 4	Arrow 6	Arrow 7
A.	O_2	Electrons	H_2O	RuBP
B.	H_2O	NADPH	O_2	CO_2
C.	H_2O	Electrons	CO_2	RuBP
D.	H_2O	NADPH	ATP	CO_2

22. Where is the location of **Pathway Q** in a chloroplast?

- A. Matrix of the chloroplast.
- B. Stroma of the chloroplast
- C. Cisternae of the chloroplast
- D. Thylakoid membrane of the chloroplast.

23. To which of the arrows in the figure below do the following statements apply?



- I. NAD^+ is regenerated without the use of the electron transport system.
- II. ATP is synthesized via substrate level phosphorylation.
- III. It can take place under aerobic conditions.

	I	II	III
A.	Q	R	T
B.	P	R	T
C.	T	Q	P
D.	T	Q	R

24. The following observations refer to evolution.

1. There is a struggle for existence.
2. Although populations tend to overproduce, they remain more or less constant in numbers from generation to generation.
3. Inherited variations which are 'favoured' in a particular environment are passed on.
4. In time, 'favoured' inherited variations may accumulate, causing gradual changes in the species.

In which sequence should they be placed to support Darwin's theory of evolution by natural selection?

- A. 1, 2, 3 and 4
- B. 1, 2, 4 and 3
- C. 2, 1, 3 and 4
- D. 2, 3, 4 and 1

25. The nucleotide sequence of the protein **TUB-1** from P were compared to **TUB-1** protein isolated from organisms Q, R, S and T to determine their evolutionary history. The DNA sequences are as follows:

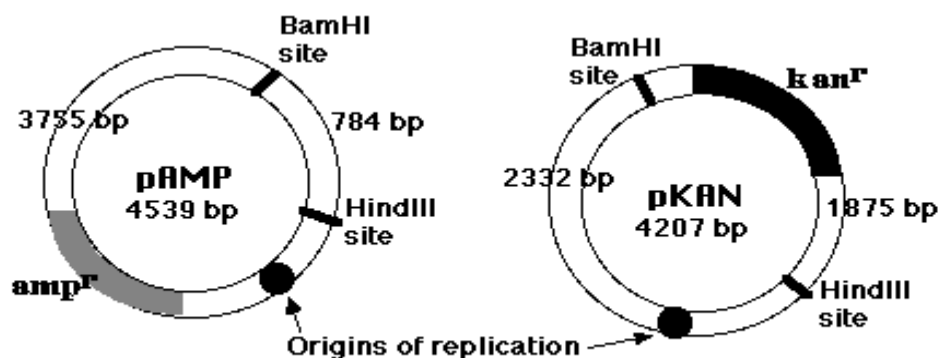
DNA sequence for TUB-1

Organism P	AGGGTTTCCCGGTTTAA AGCCTTTGGGG CCACCATTGTTGGCA
Organism Q	AGGGTTTGGCGGTTTAT AGCCTTTGCCG GGACCATTGCCCA
Organism R	AGGGTTTCC CCGTTTAAA GGCTTTGGGGCCACCATTGTTGGCA
Organism S	AGGGTTTCCGGGTTTATA GCCTTTGGGGCCCT CCATTGTTGGCA
Organism T	AGGGTTTCCCGC TTAAACCCTTTGGGGCCCTCC ATTTGGCA

Which organism is the most distant relative of **P**?

- A. Q
 - B. R
 - C. S
 - D. T
28. Which of the following is not one of the objectives of the Human Genome Project?
- A. Create a detailed genetic map of every human chromosome.
 - B. Clone human beings.
 - C. Determine the sequence of all expressed human genes.
 - D. Determine the complete DNA sequence of each human chromosome.

Refer to the diagram below for questions 26 and 27.



Two types of plasmids, pAMP and pKAN are illustrated above. pAMP carries an ampicillin-resistant gene (amp^r) and pKAN carries a kanamycin-resistant gene (kan^r).

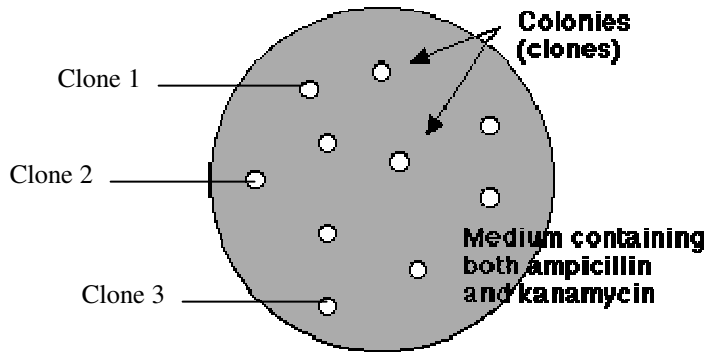
26. Solutions of these two plasmids are mixed together with the enzymes BamHI and HindIII. The pAMP and pKAN fragments were then allowed to ligate in the presence of DNA ligase.

Which of the following represent reannealed molecules that could be used as a vector in DNA cloning?

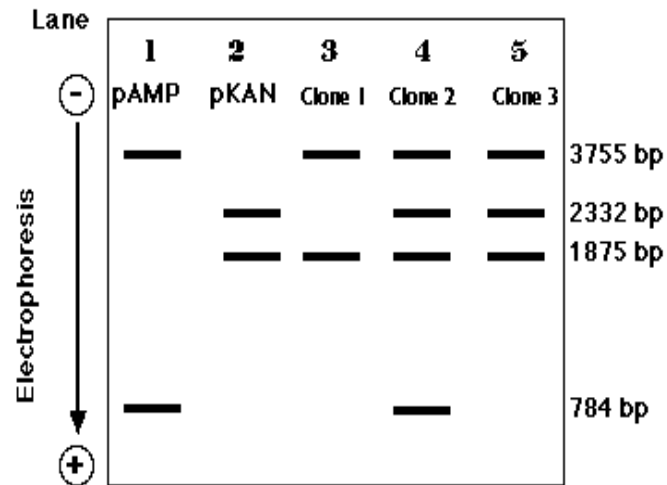
- (1) 1875 bp
- (2) 2659 bp
- (3) 3116 bp
- (4) 5630 bp

- A. 4 only
- B. 1 and 3
- C. 3 and 4
- D. 2, 3 and 4

27. Treatment of *E. coli* with the mixture of religated molecules formed in **Question 26** produces some colonies that are able to grow in the presence of both ampicillin and kanamycin.



Plasmid from clones 1, 2 and 3 were isolated and treated with enzymes BamHI and HindIII. Electrophoresis was carried out on these samples with results as shown in the diagram below.



Which of the following statements are true?

- 1 Clone 1 is transformed by a recombinant plasmid
- 2 Clone 2 is transformed by both religated pKAN and religated pAMP.
- 3 Clone 3 is transformed by the recombinant plasmid as well as the religated pKan

- A. 1 only
- B. 2 only
- C. 1 and 2
- D. 1, 2 and 3

29. Which feature of stem cells obtained from blood in the bone marrow enables their use in the treatment of a variety of blood cancers?

- A. They are totipotent.
- B. They are pluripotent.
- C. They are multipotent.
- D. They are unipotent.

30. Plants are more readily manipulated by genetic engineering than are animal cells because

- A. plant genes do not contain introns.
- B. a somatic plant cell can often give rise to a complete plant.
- C. plant cells have larger nuclei.
- D. more vectors are available for transferring recombinant DNA into plant cells.